



Pharmacy Management System

SE2012 –Object-Oriented Design & Analysis

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Introduction and Project Overview

This report is about creating a Pharmacy Management System for a real client as part of our SE2012 – Object Oriented Design and Analysis subject. The system is being built using Java Spring Boot and follows object-oriented programming concepts.

The main goal of the system is to help a pharmacy manage its work more easily.

Customers can look for medicines, place orders, and upload prescriptions.

Pharmacists can add or update medicine details, check stock, and approve or reject orders with prescriptions. Admins can manage users, view reports, and check system activities.

Our system has the following main parts:

- User Management – Register, login, and control user roles
- Inventory Management – Add, update, and monitor medicine stock
- Order and Prescription Handling – Customers can place orders and upload prescriptions
- Dashboard and Reports – Admin can see sales and inventory reports.

This system is designed to be simple to use, work on both computers and mobile phones, and keeps user data safe with secure login and role-based access. All four team members work on both the front-end and back-end equally and use a shared GitHub repository for development.

Client Details

- Client Name: A.M. Lilan Kanishka
- Business Name: New Central Pharmacy
- Location: No. 125/F, Attangalla Road, Wathupitiwala
- Business Type: Retail Pharmacy
- Contact Number: 077-7521954 / 033-2881750
- Email Address: lkmartyn@gmail.com

The client requested a system that can help manage stock levels, process customer orders efficiently, and generate sales and inventory reports. Meetings were held on

March 5, 2025 and March 9, 2025 to gather functional and non-functional requirements directly from the client.

Functional Requirements

The pharmacy website will enable pharmacists, administrators, and customers to manage medicines, prescriptions, orders and reports efficiently. Here's the functional requirement.

1.Medicine Management

- Easily add, update, or remove medicines: Pharmacists and admins can add new medicines, update details like price or expiry date or remove items that are no longer in stock. This helps keep the inventory accurate and up to date.
- Automatic stock and expiry monitoring: The system will keep track of stock levels and expiry dates. When medicines are running low or about to expire, the system will alert the relevant staff.

2.User Management

- Registration and secure login: Customers can create their own accounts, while admins can set up accounts for pharmacists. Everyone can log in and log out securely.
- Manage roles and permissions: The system shall enable admins to assign and manage user roles.

3.Prescription and Order Management

- Place orders for medicines: Customers can purchase over-the-counter medicines directly, while prescription-only medicines require uploading a valid prescription and the order will be managed by a pharmacist.
- Order processing and status tracking: Pharmacists review and approve prescription orders, while all customers can track the status of their orders at any time.
- Order history and cancellation: Customers can view their past orders and cancel orders if needed, giving them more control and transparency.

4. Pharmacist Dashboard

- View and manage medicine inventory: Pharmacists can easily view current inventory, update stock information, and manage medicine details from their dashboard.
- Receive low stock and expiry date alerts: The system provides real-time alerts for medicines that are running low or nearing expiry, enabling pharmacists to take timely action.
- View and process prescription orders: Pharmacists can review incoming prescription orders, verify uploaded prescriptions, and process or update order statuses efficiently.

Non-Functional Requirements

1. Security

- Only authorized users can access sensitive features like inventory and reports.
- Username and User passwords are securely stored in the database.

2. Usability

- The interface is designed to be user-friendly and easy to navigate.
- The system will guide users clearly through all actions like placing an order or managing inventory.

Use Case Diagram (Group)



Figure 1 - Use Case Diagram (Group)

Class diagram (Group)

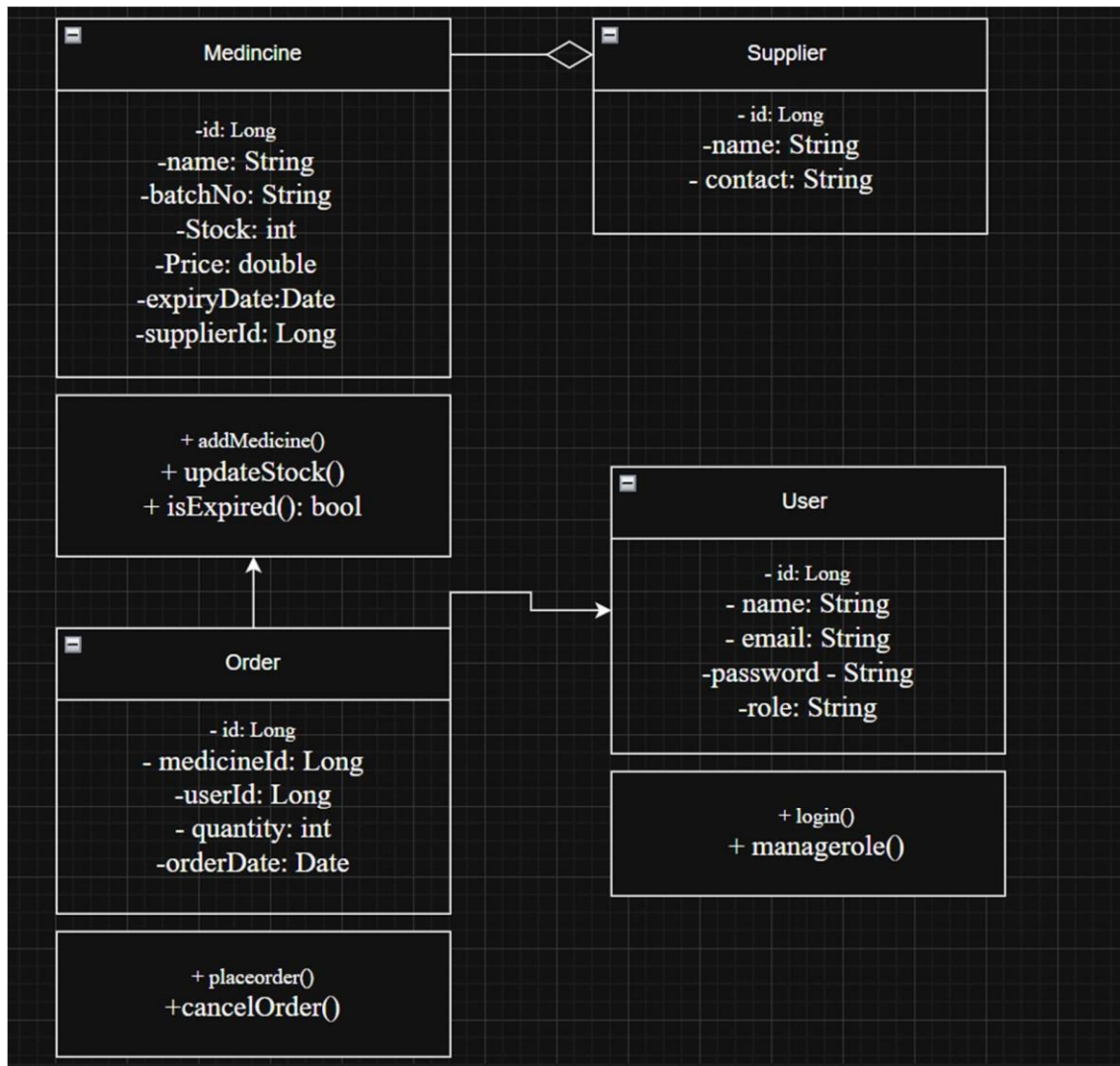


Figure 2 - Class Diagram (Group)

User case scenarios and Activity Diagrams

User Management use case scenario and Activity Diagram- SAMPATH L A S S S

Use Case 1: Registering a New User Account

- **Number:** UM001
- **Name:** Registering a New User Account
- **Summary:** Customer creates a new account to access pharmacy services
- **Priority:** 1
- **Pre-condition:** Customer is not currently registered
- **Post-condition:** A new user account has been created in the system
- **Primary actor:** Customer
- **Trigger:** Customer initiates registration from login page

Main Scenario

Table 1 - Registering a New User Account Main Scenario

Step	Action
1	Customer selects “Register” from login page
2	The system displays the registration form
3	Customer fills in name, email, phone, and password
4	System validates the input
5	If valid, the system saves new users in the database
6	System confirms registration and redirects to login screen

Extensions

Table 2 - Registering a New User Account Extensions

Step	Branching Action
4a	Email already in use → system displays error and asks for another
4b	Empty required field → system highlights it and prompts for completion

Use Case 2: Managing User Roles

- **Number:** UM002
- **Name:** Assigning or Updating User Roles
- **Summary:** Admin updates a user's role (Admin, Pharmacist, or Customer)
- **Priority:** 2
- **Pre-condition:** Admin is logged into the system
- **Post-condition:** User's role is updated in the system
- **Primary actor:** Admin
- **Trigger:** Admin navigates to the User Management panel

Main Scenario

Table 3 - Managing User Roles Main scenario

Step	Action
1	Admin logs into the system
2	Opens User Management interface
3	Searches for a user by email or ID
4	Selects the user and chooses a new role
5	System updates the user's role
6	Confirmation message is displayed

Extensions

Table 4 - Managing User Roles Extensions

Step	Branching Action
3a	User not found → system displays error
5a	Role not selected → system prompts selection

Use Case 3: Deleting a User Account

- **Number:** UM003
- **Name:** Removing a User Account
- **Summary:** Admin deletes an account from the system
- **Priority:** 2
- **Pre-condition:** Admin is logged into the system
- **Post-condition:** Selected user accounts are removed from the system
- **Primary actor:** Admin
- **Trigger:** Admin initiates delete action on a selected user

Main Scenario

Table 5 - Deleting a User Account Main scenario

Step	Action
1	Admin opens User Management
2	Searches and selects the user to delete
3	Clicks “Delete” and confirms action
4	System removes the user account
5	System shows success message

Extensions

Table 6 - Deleting User Account Extensions

Step	Branching Action
2a	User not found → error displayed
3a	User is a protected Admin → deletion denied

Activity Diagram of User Management part

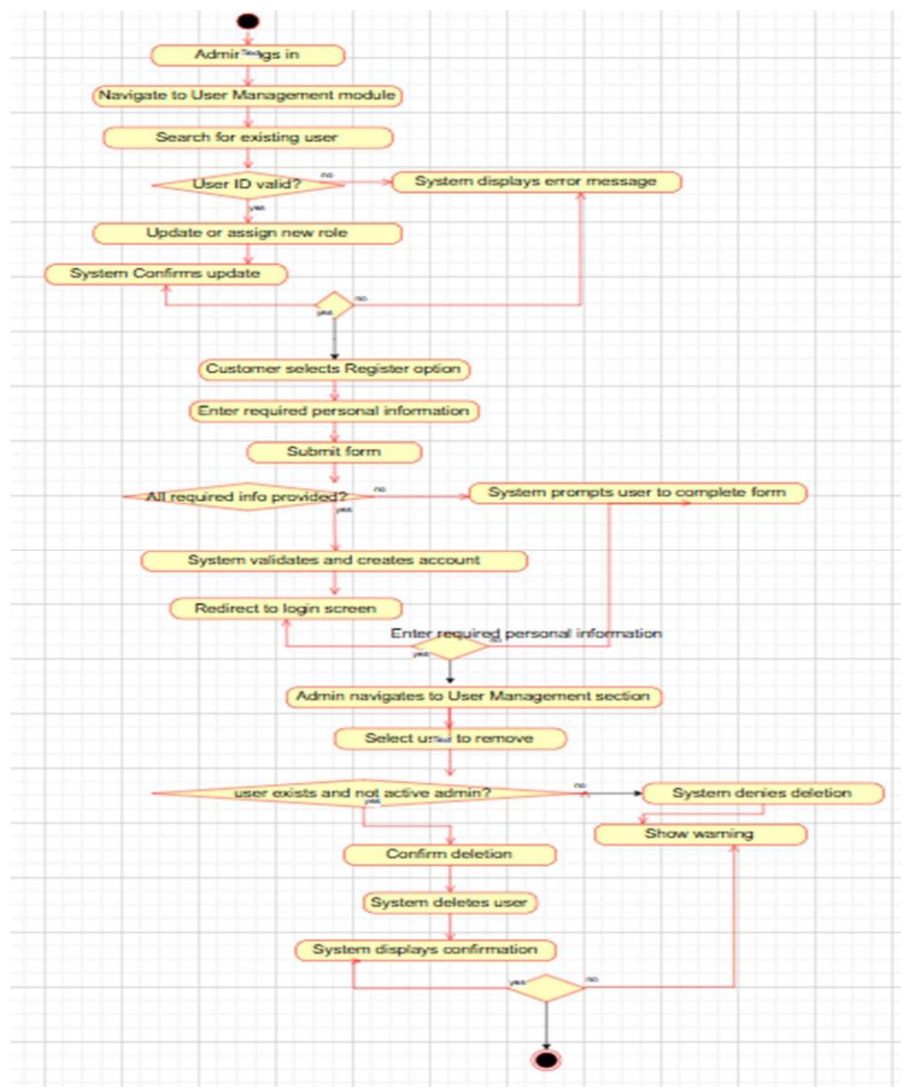


Figure 3 - Activity Diagram of User Management part

Inventory Management User case scenario and Activity Diagram (Anojan.R)

Pre-Conditions

- Pharmacists must be logged into the system.
- The inventory module must be accessible and connected to the database.
- Medicines and stock data are available in the database.

Post-Conditions

- Pharmacist successfully adds, updates, or deletes medicine.
- System stores updated inventory data in the database.
- The inventory list is refreshed with new values.

Table 7 - Inventory Management main scenario

Step	Action
01	Pharmacist logs into the system using valid credentials
02	Pharmacist navigates to the Inventory Management section
03	System displays options: Add, Update, Delete, View Medicines
04	Pharmacist selects Add New Medicine

05	System prompts for medicine details (name, batch no, stock, expiry date, etc.)
06	Pharmacist fills the form and submits
07	System validates the input and saves medicine to the database
08	System displays success message and refreshes inventory list

Main Scenario

This use case describes how a pharmacist manages medicine inventory in the system. It involves logging in, selecting inventory actions (add/update/delete/view), and ensuring data is saved successfully

Extensions

Table 8 - Inventory Management Extensions

Step	Branching Action
05a	If required fields are missing → System prompts to complete the form
07a	If duplicate batch number exists → System shows "Duplicate Batch" warning
07b	If database error occurs → System displays error and prevents submission

Activity Diagram of Inventory Management part

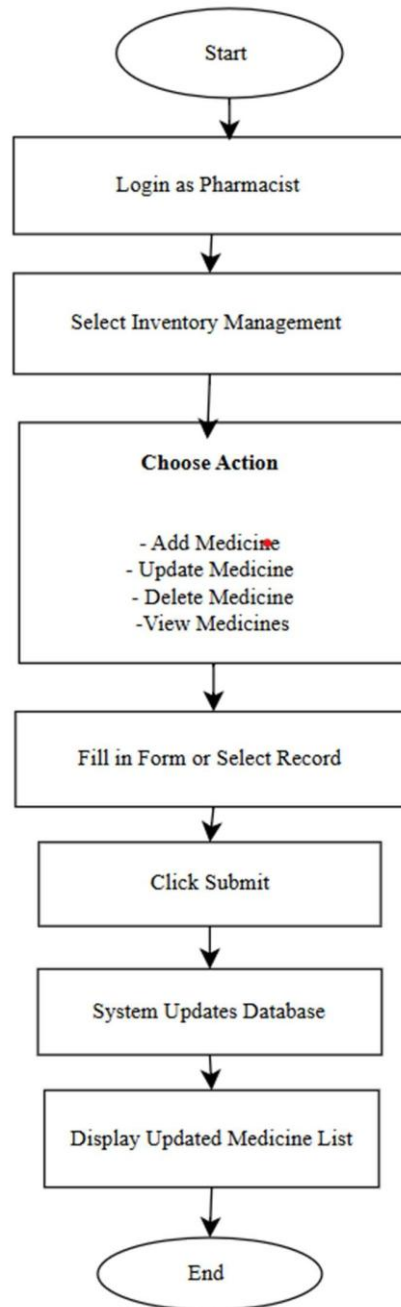


Figure 4 - Activity Diagram of Inventory Management part

Customer Places an Order Use Case Scenario and Activity Diagram (K.M.T.P.Kulasinha)

- Priority – High
- **Pre-Condition** – Customer must log into the system
- **Post-Condition** - Order is saved and visible in Order History; status is set to “Pending Approval” if prescription is uploaded

Main Scenario

Table 9 - Customer Places an Order main scenario

Step	Action
1	Customer logs into the system
2	Customer browses available medicines
3	Customer adds medicines to the cart
4	System checks if any selected item requires a prescription
5	If required, system prompts customer to upload prescription
6	Customer uploads prescription
7	Customer reviews and confirms the order
8	System saves the order and sets order status to “Pending Approval”

9	System shows confirmation and order appears in customer's order history
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Extensions

Table 10 - Customer Places an Order Extensions

Step	Branching Action
1a	Invalid login – system asks to re-enter login details
6a	No prescription uploaded – system prevents order confirmation and prompts again

Activity diagram of Customer Places an Order part

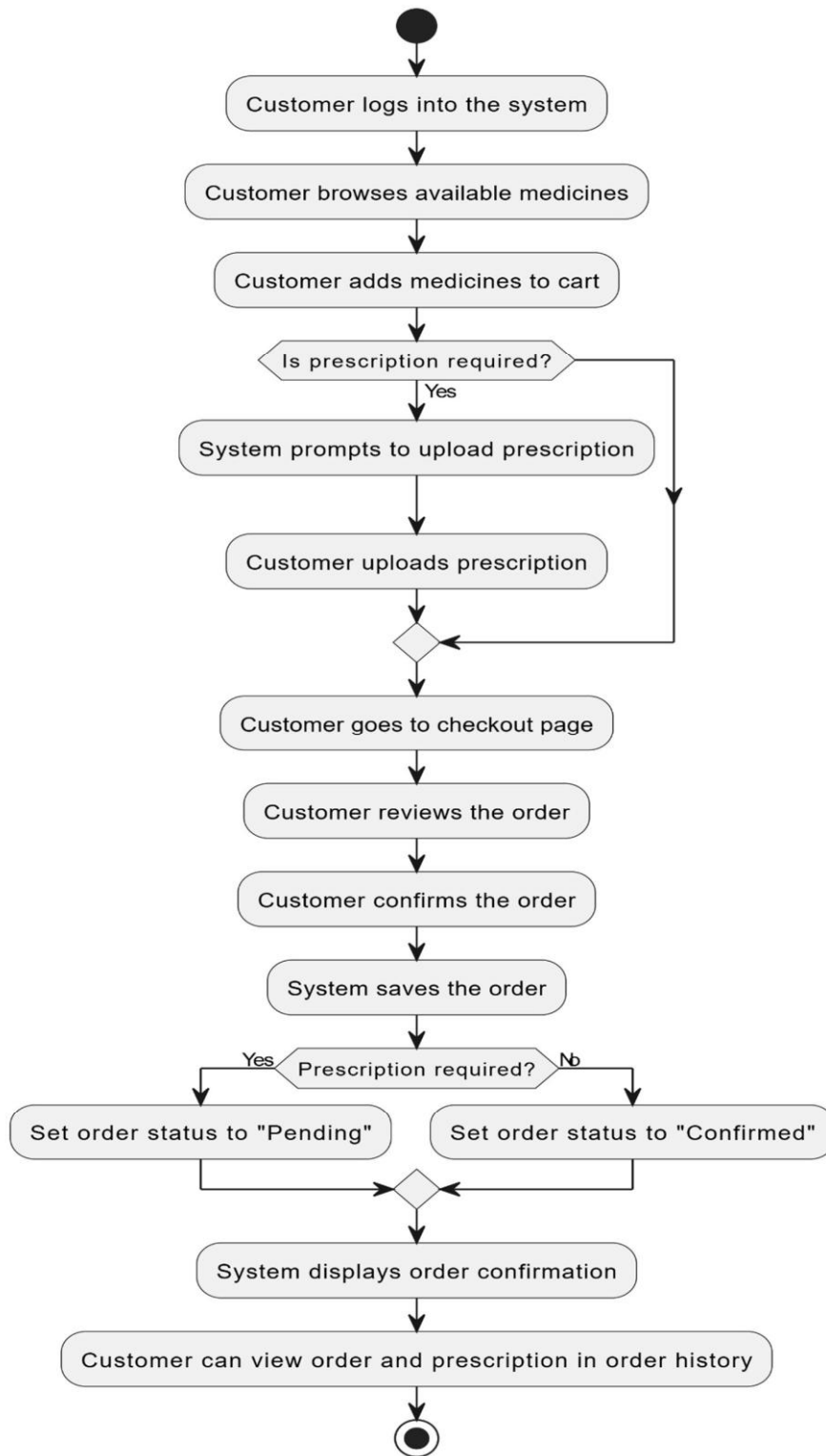


Figure 5 - Activity Diagram of Customer Places an Order part

Admin Dashboard & Reports Use Case Scenario and Activity Diagram (DISSANAYAKE D M J T)

Pre-Conditions

- Users must be logged into the system.
- Sales, inventory, and user role data are available in the database.
- The system is operational and connected.

Post-Conditions

- Pharmacist completes viewing reports and monitoring alerts.
- Admin updates or creates user roles with the appropriate permissions.

Main Scenario

Table 11 - Admin Dashboard & Reports main scenario

Step	Action
01	User logs into the system using valid credentials
02	Pharmacist Selects “View Reports”
03	System prompts to select a report type
04	Pharmacist selects Sales → System displays Sales Report
05	Pharmacist selects Inventory → System displays Inventory Report
06	Pharmacist navigates to Alerts
07	Pharmacist selects Low Stock Alerts → System displays medicines with low stock

08	Pharmacist selects Expiry Date Alerts → System displays medicines nearing expiry
09	Admin selects User Management
10	System checks if the role exists
11	If role exists → Admin edits role permissions
12	If role doesn't exist → Admin creates a new role with specific permissions

Extensions

Table 12 - Admin Dashboard & Reports Extensions

Step	Branching Action
3a	No report type selected → System prompts to choose a report type
10a	Role already exists → System prevents duplication and displays a message

Activity Diagram of Admin Dashboard & Reports part.

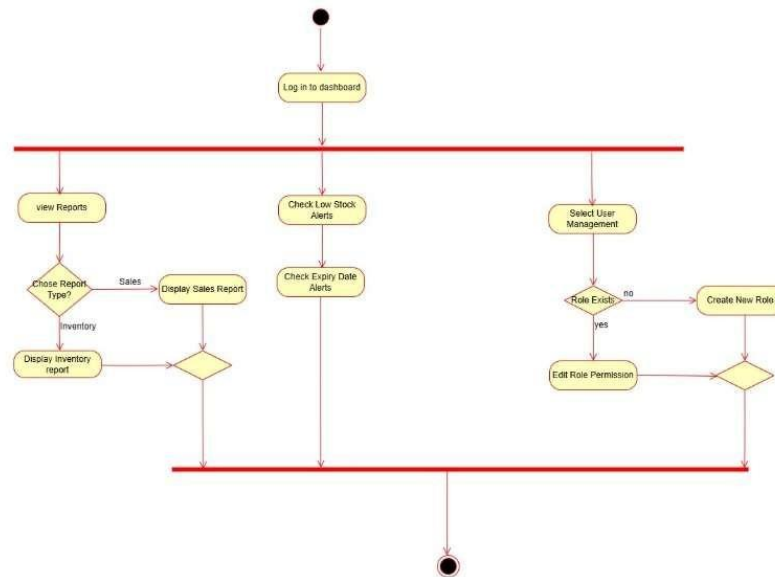


Figure 6 - Activity Diagram of Admin Dashboard & Reports part

Challenges Faced

1. Technical Problems

During the development of the Pharmacy Management System, we encountered several technical difficulties, especially when setting up the backend using Spring Boot. Initially, configuring dependencies and connecting the backend to the MySQL database caused delays due to version mismatches and unfamiliarity with Spring JPA. Another issue was integrating the frontend with the backend using RESTful APIs, particularly handling CORS errors and managing data flow between components.

To overcome these, we referred to official documentation, used community forums like Stack Overflow, and divided the research tasks among members. We also

simplified our API responses and used tools like Postman to test and debug each endpoint individually before connecting it to the frontend.

2. Group Coordination & Time Management

As this was a group project with different responsibilities, coordinating tasks and maintaining consistent progress was sometimes difficult. Some members had overlapping assignments or exam schedules, which occasionally led to delays in feedback or code sharing.

We resolved this by scheduling weekly online meetings, creating a shared project folder with clear subfolders for frontend, backend, and documentation, and assigning deadlines using a shared task list. This ensured better accountability and allowed everyone to track progress. Communication through WhatsApp and Google Meet helped us clarify issues quickly and collaborate more effectively.

3. Learning Curve

Team members were new to technologies like Spring Boot, React, or database management. As a result, time had to be allocated for learning and experimenting before actual development could begin.

We tackled this by watching beginner-friendly tutorials on YouTube, using hands-on practice projects, and sharing helpful resources among group members. This not only solved the issue but also improved everyone's skills and confidence in full-stack development.

Conclusion

The development of the Pharmacy Management System provided us with a valuable opportunity to apply object-oriented design principles and full-stack development skills in a real-world context. Through this project, we successfully built a system that allows pharmacies to manage their inventory, user accounts, orders, and reporting tasks efficiently.

What Was Achieved

We implemented a web-based system using Spring Boot for the backend, MySQL for data storage, and React/HTML for the frontend interface. Each group member took

responsibility for one key module — User Management, Inventory Management, Order Processing, and Admin Dashboard & Reports — and ensured smooth integration of all components.

We were able to:

- Build working APIs for CRUD operations and business logic
- Connect backend with frontend using RESTful services
- Implement role-based access control for security
- Create diagrams, wireframes, and documentation that support our system design

Key Takeaways

- Gained hands-on experience with full-stack technologies
- Learned how to structure and manage a group software project
- Improved problem-solving skills, especially while debugging code and configuring frameworks
- Understood the importance of proper planning, regular communication, and task distribution

Final Remarks on Teamwork & Learning

This project helped us understand the importance of team collaboration, version control, and consistent communication. By dividing responsibilities and supporting each other, we were able to complete the project efficiently despite challenges. Most importantly, it boosted our confidence to take on larger software engineering tasks in future academic or professional settings.

References

1. Spring Boot Documentation - <https://spring.io/projects/spring-boot>
2. Spring Data JPA Documentation - <https://docs.spring.io/spring-data/jpa/docs/current/reference/html/>
3. MySQL Reference Manual - <https://dev.mysql.com/doc/>
4. Draw.io – UML & ER Diagram Tool - <https://app.diagrams.net/>
5. GeeksforGeeks – Java and Spring Boot Tutorials - <https://www.geeksforgeeks.org/>

6. W3Schools – HTML, CSS, JavaScript Basics - <https://www.w3schools.com/>
7. YouTube – Spring Boot & Full Stack Projects
8. React Documentation - <https://reactjs.org/docs/getting-started.html>
9. Stack Overflow – Troubleshooting and code help - <https://stackoverflow.com/>
10. ChatGPT & Grok AI

Appendix

Meeting 1 – Initial Discussion (Physical)

Date: March 05, 2025

Objective: Understand the needs of the pharmacy

Notes:

- Client requested an online inventory system
- Wants expiry date tracking and stock alerts
- Admin should be able to generate reports
- The system should be user-friendly for non-technical staff



Figure 7 – Initial Discussion Image

Meeting 2 – Online Meeting

Date: March 13, 2025

Objective: More requirements gathered

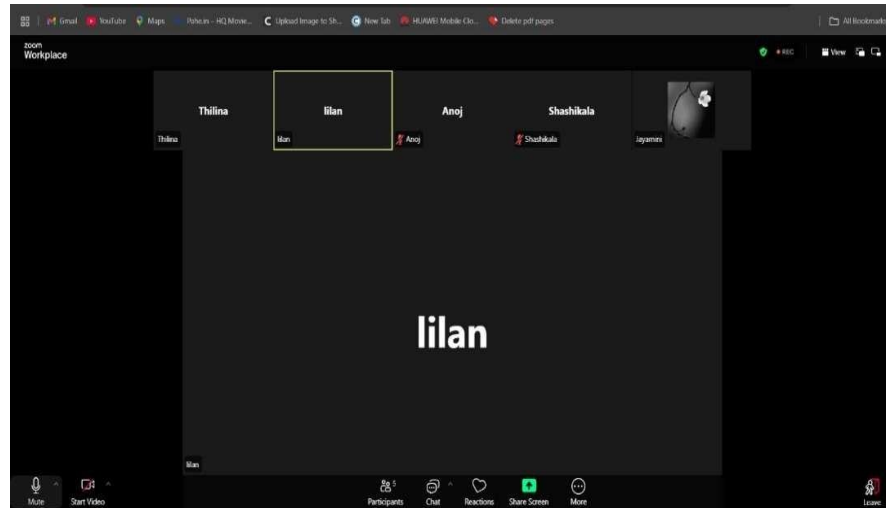


Figure 8 - Online Meeting Screenshot

Client Offer letter

A.M. Lilan Kanishka
New Central Pharmacy
No 125/F, Attangalla Road, Wathupitiwala.
077-7521954 / 033-2881750
2025-03-01

Request for a Pharmacy Management System

Dear team,

I am writing to formally express my interest in implementing a Pharmacy Management System for New Central Pharmacy. Currently, our manual processes are leading to several challenges, including difficulty in tracking stock levels and expiry dates, inefficiency in managing prescriptions, and the absence of automated reports for business insights.

To address these challenges, I require a system with the following features:

- **Automated Inventory Tracking** – To monitor stock levels, receive low-stock alerts, and manage expiry dates.
- **Prescription & Order Management** – To store and verify prescriptions while streamlining the order processing system.
- **User Management** – Role-based access for pharmacists, administrators, and customers.
- **Admin Dashboard & Reporting** – A real-time reporting system for tracking sales, stock levels, and other critical analytics.

I look forward to collaborating with your team to develop a solution that meets my business requirements. Please provide an estimated timeline, budget, and any further information required to proceed.

Thank you for your time and consideration. I am looking forward to your response.

Sincerely,


A.M. Lilan Kanishka
New Central Pharmacy

Figure 9 - Client Offer letter